

How to remove the seasonal atmospheric effect: An application on a surface temperature at Nisyros volcanic monitoring station, Hellenic Volcanic Arc.

Abstract

A surface temperature at a volcanic monitoring system is the result of a mixture of two heat sources originating one from above and another from below the ground surface. The atmosphere significantly affects the surface's temperature of the volcano, and removal of the atmospheric effect is required for meaningful interpretations and reliable volcanic hazard evaluation. Time series of meteorological data were acquired from the closest weather station to Nisyros volcano, and surface volcanic temperature from the southern part of Nisyros caldera were analyzed. Time series analysis took place at different time scales through a decomposition method. The decomposition of the time series into long-term, seasonal-term, and short-term components was performed by the application of the Kolmogorov-Zurbenko (KZ) filter to separate the noise that appears on the obtained data. The Time Series Regression (TSR) model was applied in all time scales. The long- and the short-term components of the surface volcanic temperature show no significant correlation with the atmospheric variables, while the seasonal term indicates a very strong association with an R2 equal with 98.7%. This methodology may facilitate the process of filtering a seasonal atmospheric effect and better comprehend the heat emission and related volcanic hazard at a volcano-hydrothermal system from temperature surface readings.

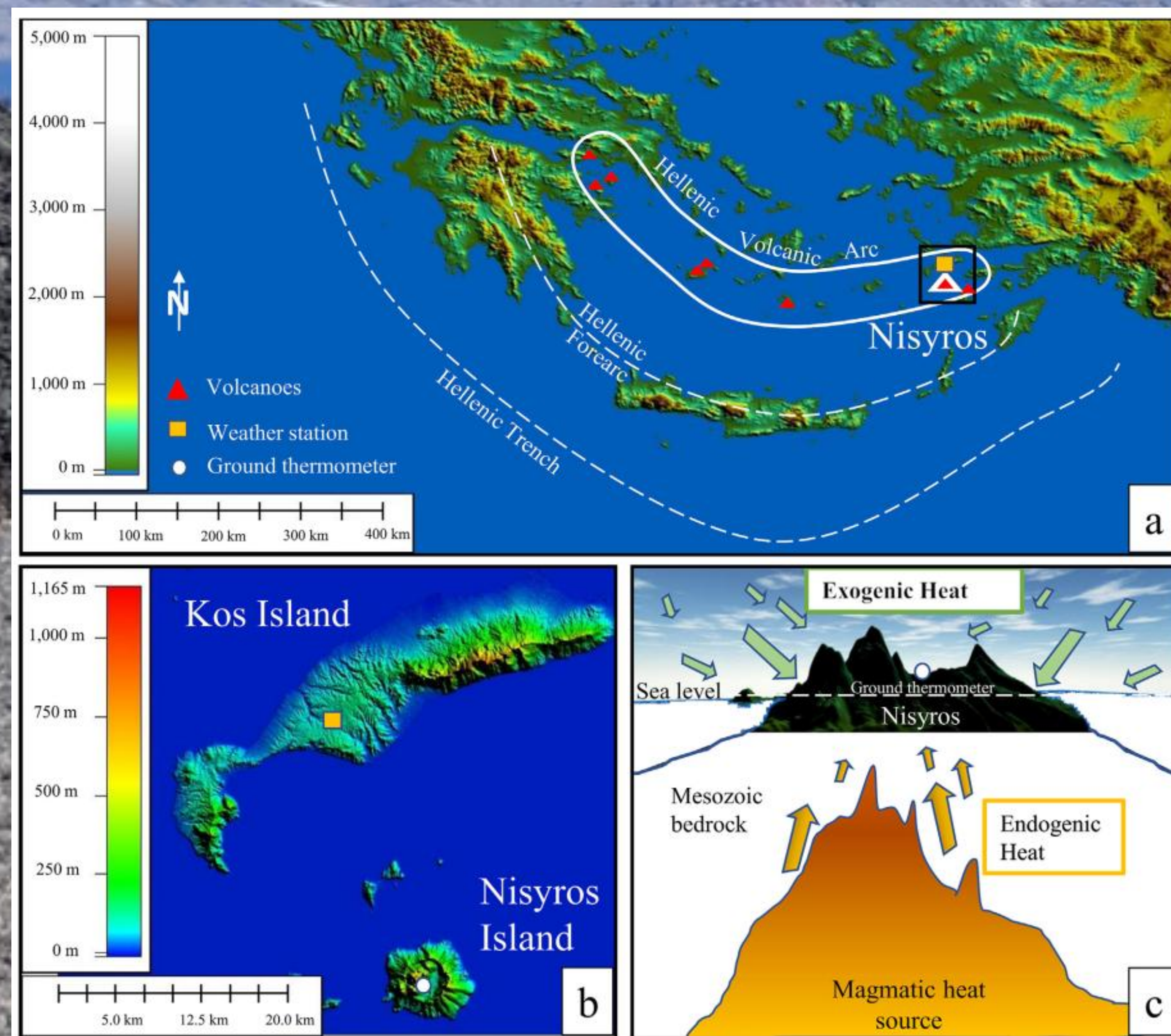


Figure 1: (Top) Location of the study area. (Bottom left): Digital elevation model of the surrounding area. (Bottom right): A conceptual figure of the heat contribution to the surface temperature.

Introduction

- A surface temperature at a volcanic monitoring system is the result of a mixture of two heat sources (Fig. 1) originating one from above and another from below the ground surface.
- The atmosphere significantly affects the surface's temperature of the volcano, and removal of the atmospheric effect is required for meaningful interpretations and reliable volcanic hazard evaluation.

Methods

Collecting Fumarole Temperature Data

- Time series of meteorological data were acquired from the closest weather station to Nisyros volcano, and surface volcanic temperature from the southern part of Nisyros caldera were analyzed.
- Time series analysis performed in R programming language took place at different time scales through a decomposition method. The decomposition of the time series into long-term, seasonal-term, and short-term components was performed by the application of the Kolmogorov-Zurbenko (KZ) filter to separate the noise that appears on the obtained data (Fig.2).
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- The Time Series Regression (TSR) model was applied in all time scales.

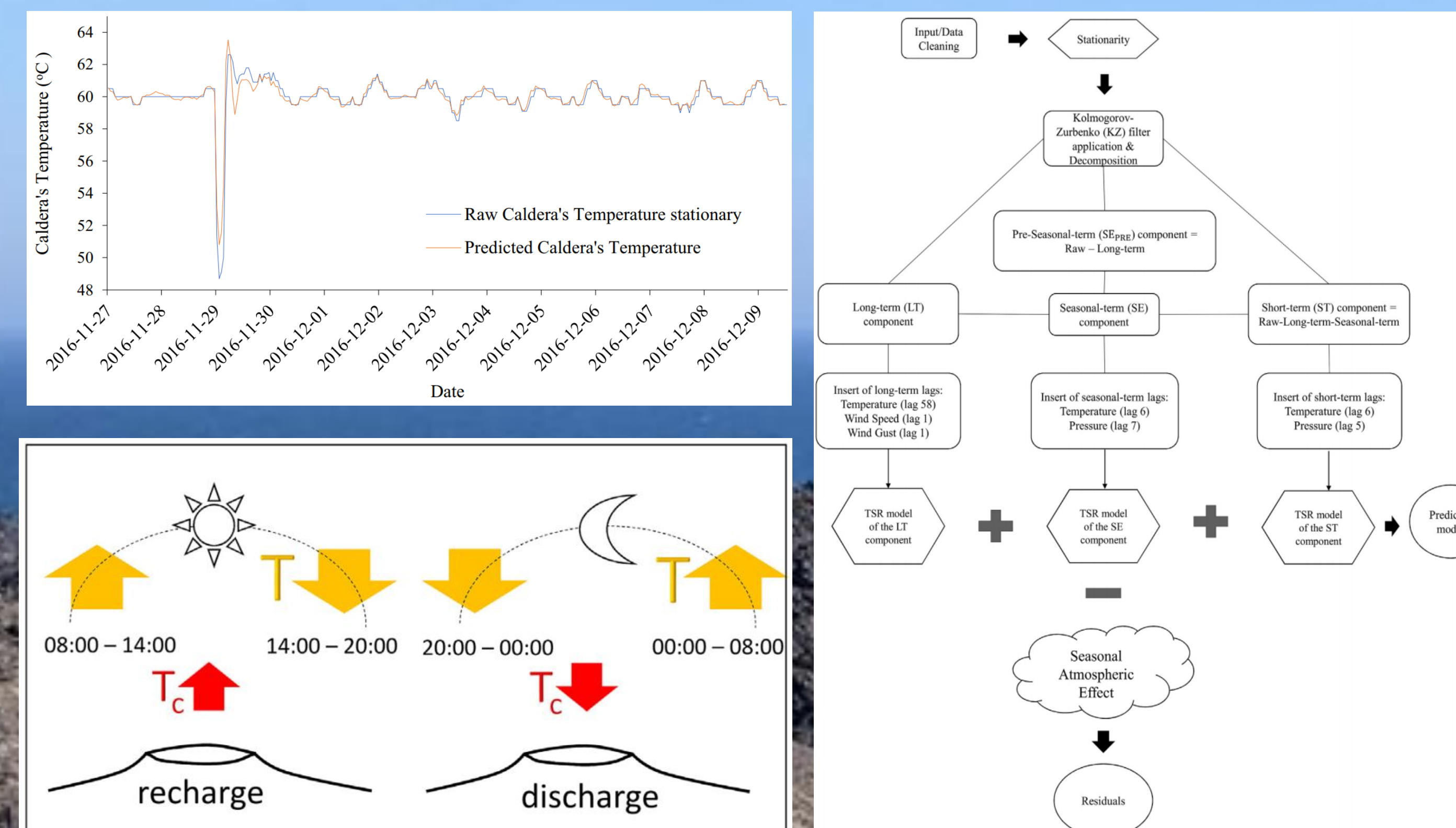


Figure 2: (Top right) Data process flowchart of the prediction model. (Top left): Prediction model against the real temperature data. (Bottom left): A conceptual figure of the heat contribution to the surface temperature and the related charging cycles of the surface.

Results

- The long- and the short-term components of the surface volcanic temperature show no significant correlation with the atmospheric variables, while the seasonal term indicates a very strong association with an R2 equal with 98.7%.
- Seasonal atmospheric influence is strong and it masks other significant frequencies (Fig. 3).

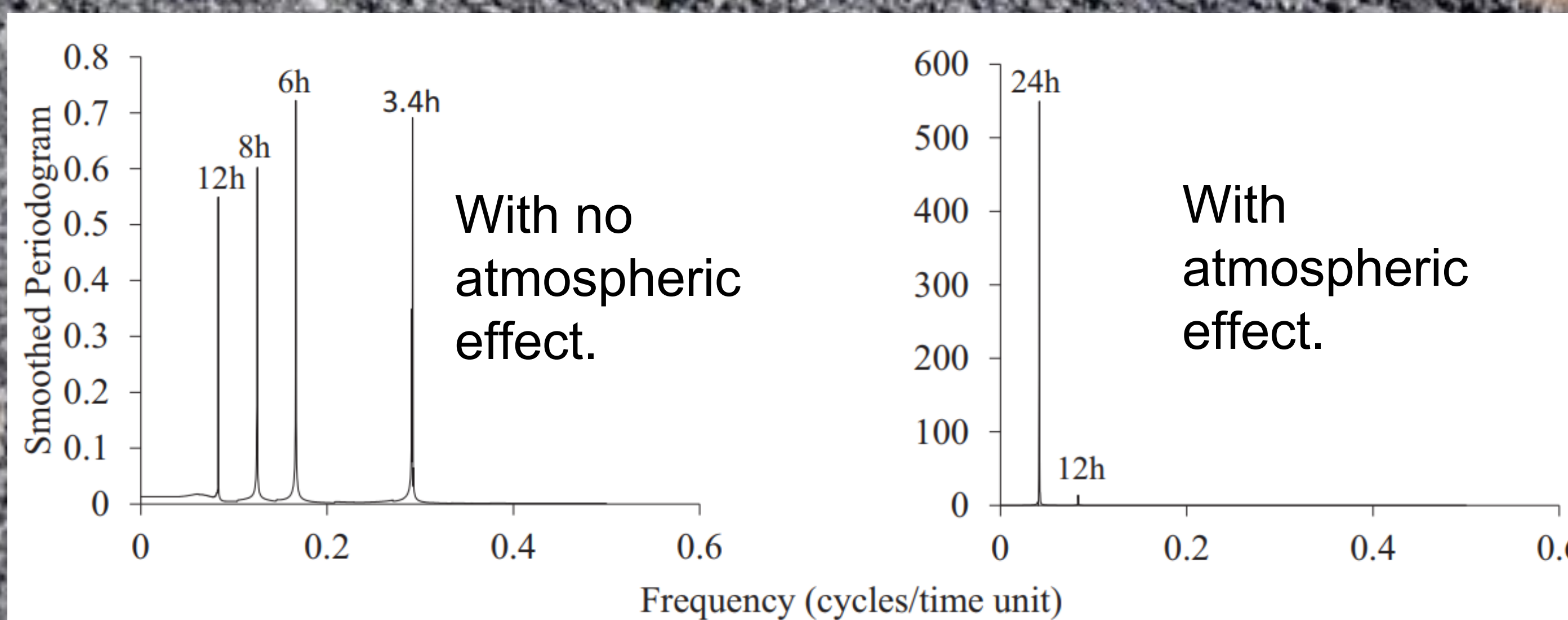


Figure 5: The main periodicities of the seasonal-term component of the caldera's temperature after removing the atmospheric effect (left graph) and without removal of the atmospheric influence (right graph).

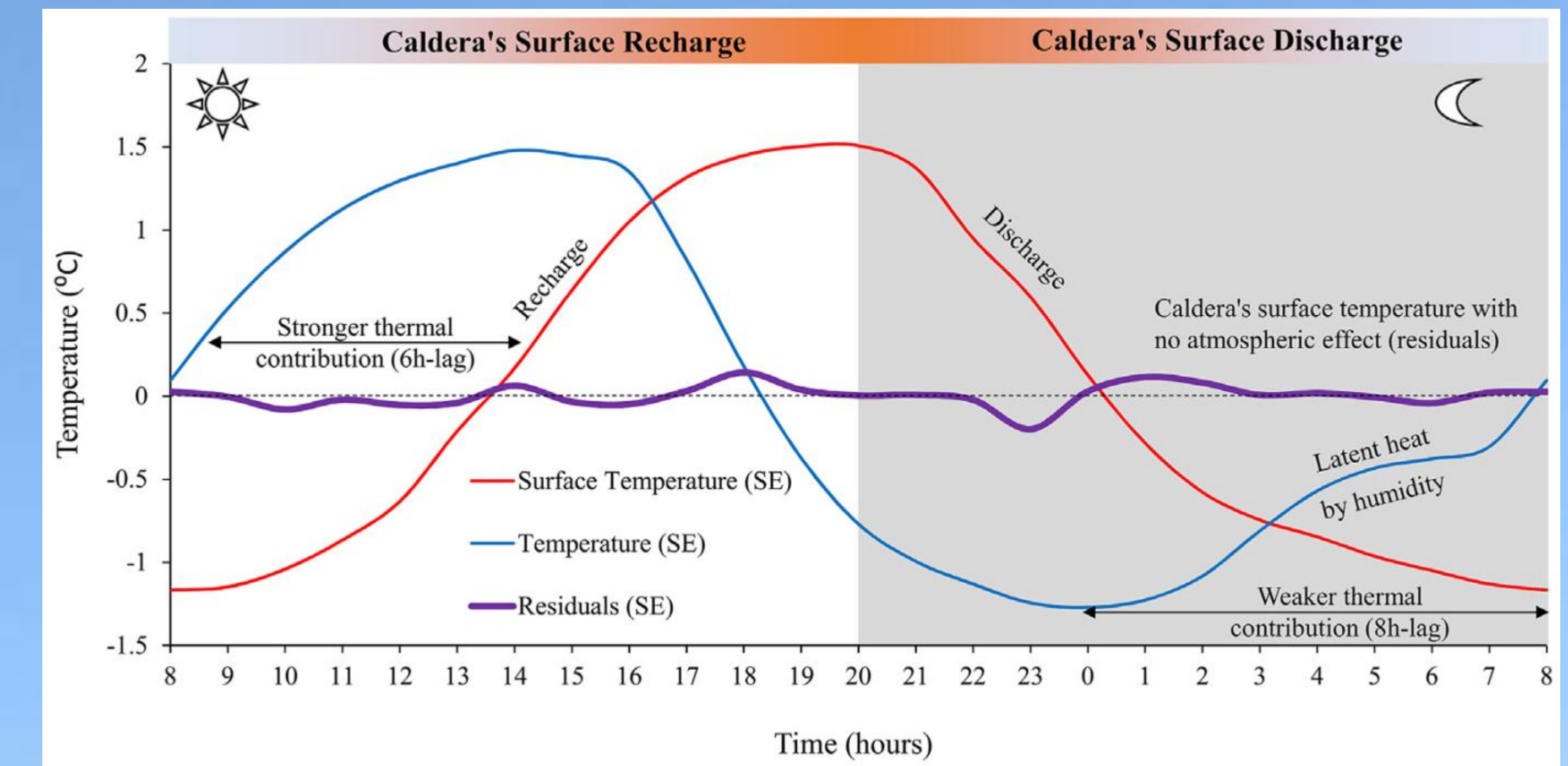


Figure 4: The cyclical phenomena of the caldera's surface recharge, discharge, and the influence of the thermal contribution on these processes from the seasonal component.

Discussion

- Unlike high temperature volcanoes, low temperature volcanoes are possibly more influenced by things like atmosphere and monthly cycles, as supported by our study. However, other factors like sea level, 11 year cycles/ extraterrestrial patterns, tectonics, or other factors could also impact the volcano's behavior.
- Atmospheric or daily temperature fluctuations could be due to heat from the sun being retained in the soil, or because the fumarole itself retains heat present during the day.
- Liquids such as seawater and hydrothermal fluids residing below the surface and above the magma chamber could be affected by the moon's gravitational pull. Pulling these liquids closer to the surface can cause an increase in fumarole temperatures, but this increase does not reflect the true temperature of the magma chamber. So, this influence is considered noise and removed through the kz function.

Conclusion

- Removal of atmospheric effect reveals hidden frequencies of thermal signature.
- The volcano showed a baseline temperature fluctuations with all influences removed, suggesting minimal volcanic activity.
- Surficial temperature manifestations may reveal deep processes and, many times, be evaluated as a risk-assessment trigger mechanism for public and scientific alerts.
- An efficient removal of the atmospheric noise from surface loggers ensures reliable interpretations.

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References

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